COMPUTING THE DERIVATION THETA ON THE TATE-RESOLUTION CLASS

THE STACKS PROJECT AUTHORS

Editorial note. No omission markers in-span; proof complete as written.

Allowed standard prerequisites (editorial).

- Derivation from a square-zero extension. lemma-get-derivation (same chapter, named; NOT carded – build-gate exclusion). Supplies the derivation theta: A/IA -> A/IA and the lifting procedure for A(i) that this proof reuses; declared as an external reference, not re-proved (its construction step is omitted in the source).
- Tate resolution. lemma-tate-resolution (same chapter, named). Supplies the filtration A(0) subset A(1) subset ... used to place the computation.

Source reference. Original lemma 09PS: sources/stacks-dpa/pinned-original-source.tex, source lines 1651–1651. Navigation pointer only: the counted proof cites this statement by \ref as a standing input; the deduction the counted proof has to make is carried out inside the counted span, and no part of that deduction is delegated to this pointer.

Lemma 0.1. *Assumption and notation as in Lemma 09PS. Suppose $S = H_0(A)$ is isomorphic to $R[x_1, \ldots, x_n]/(f_1, \ldots, f_m)$ for some n, m, and $f_j \in R[x_1, \ldots, x_n]$. Moreover, suppose given a relation*

$$\sum r_j f_j = 0$$

with $r_j \in R[x_1, \ldots, x_n]$. Choose $r'_j, f'_j \in R'[x_1, \ldots, x_n]$ lifting r_j, f_j. Write $\sum r'_j f'_j = gf$ for some $g \in R/I[x_1, \ldots, x_n]$. If $H_1(A) = 0$ and all the coefficients of each r_j are in I, then there exists an element $\xi \in H_2(A/IA)$ such that $\theta(\xi) = g$ in S/IS.

Proof. Let $A(0) \subset A(1) \subset A(2) \subset \ldots$ be the filtration of A such that $A(m)$ is gotten from $A(m-1)$ by adjoining divided power variables of degree m. Then $A(0)$ is a polynomial algebra over R equipped with an R-algebra surjection $A(0) \to S$. Thus we can choose a map

$$\varphi : R[x_1, \ldots, x_n] \to A(0)$$

lifting the augmentations to S. Next, $A(1) = A(0)\langle T_1, \ldots, T_t\rangle$ for some divided power variables T_i of degree 1. Since $H_0(A) = S$ we can pick $\xi_j \in \sum A(0)T_i$ with $\mathrm{d}(\xi_j) = \varphi(f_j)$. Then

$$\mathrm{d}\left(\sum \varphi(r_j)\xi_j\right) = \sum \varphi(r_j)\varphi(f_j) = \sum \varphi(r_j f_j) = 0$$

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Since $H_1(A) = 0$ we can pick $\xi \in A_2$ with $\mathrm{d}(\xi) = \sum \varphi(r_j)\xi_j$. If the coefficients of r_j are in I, then the same is true for $\varphi(r_j)$. In this case $\mathrm{d}(\xi)$ dies in A_1/IA_1 and hence ξ defines a class in $H_2(A/IA)$.

The construction of θ in the proof of Lemma 09PS proceeds by successively lifting $A(i)$ to $A'(i)$ and lifting the differential d. We lift φ to $\varphi' : R'[x_1, \ldots, x_n] \to A'(0)$. Next, we have $A'(1) = A'(0)\langle T_1, \ldots, T_t\rangle$. Moreover, we can lift ξ_j to $\xi'_j \in \sum A'(0)T_i$. Then $\mathrm{d}(\xi'_j) = \varphi'(f'_j) + fa_j$ for some $a_j \in A'(0)$. Consider a lift $\xi' \in A'_2$ of ξ. Then we know that

$$\mathrm{d}(\xi') = \sum \varphi'(r'_j)\xi'_j + \sum fb_iT_i$$

for some $b_i \in A(0)$. Applying d again we find

$$\theta(\xi) = \sum \varphi'(r'_j)\varphi'(f'_j) + \sum f\varphi'(r'_j)a_j + \sum fb_i\mathrm{d}(T_i)$$

The first term gives us what we want. The second term is zero because the coefficients of r_j are in I and hence are annihilated by f. The third term maps to zero in H_0 because $\mathrm{d}(T_i)$ maps to zero. $\square$